Mathematical Theory of Finance: Assignment 1

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Problem 1

1. **Why don't we just print more money?:** Bonds give the government the power to fund themselves but also act as a stronger tool for monetary policy, allowing for spending without undermining the legitimacy of the state currency by introducing excess inflation.
2. **Yield Curve Flattening:** If long-term economic growth is uncertain but short-term is stable (possibly due to unstable political situations like new elections or instability abroad), people will bid on bonds with longer time horizons. This raises their price and brings their yield down while the short-term yield remains stable, causing the flattening.
3. **Quantitative Easing:** Quantitative easing is a monetary policy wherein the central banks increase the money supply to stimulate market growth by buying back government bonds (and other securities). For example, the government, at its peak during Covid, was purchasing "at least $80 billion a month in Treasuries and $40 billion in residential and commercial mortgage-backed securities" (Milstein & Wessel, n.d.).

Problem 2

An ideal selection of bonds should have maturity spaced roughly at 6-month intervals so that any interpolation required between points has an equal distance and doesn't magnify any errors more than others. An even 6-month distribution across 10 stocks gives us the 0–5-year yield curve we are investigating.

We want the coupon payments to be near the date of maturity of the previous stocks since that's the rate we will use to discount them. If two stocks have similar maturities, some other deciding factors are a more moderate coupon to avoid outliers, and a more recent issue date for a more accurate price. Using these as our north star, the following ten bonds have been chosen (Markets Insider, n.d.-a, Markets Insider, n.d.-b):

- `CA135087E679`: CAN 1.50 Jun 26
- `CA135087S547`: CAN 3.00 Feb 27
- `CA135087F825`: CAN 1.00 Jun 27
- `CA135087T958`: CAN 2.25 Feb 28
- `CA135087H235`: CAN 2.00 Jun 28
- `CA135087Q491`: CAN 3.25 Sep 28
- `CA135087J397`: CAN 2.25 Jun 29
- `CA135087N670`: CAN 2.25 Dec 29
- `CA135087T388`: CAN 2.75 Sep 30
- `CA135087L443`: CAN 0.50 Dec 30

Problem 3

The eigenvectors of the covariance matrix, sometimes called principal components, give us a direction of highest variance. Mathematically, when the data is projected onto this vector, it has the highest variance. Intuitively, one may think of the eigenvector as describing the mix of random variables that has most of the variation. In this context our eigenvector represents the weights of a portfolio that explains most of the variance in the market

The eigenvalues of these vectors are a measure of the variance captured by its respective eigenvector; a larger eigenvalue means that it has more variance. Ordering these eigenvectors by their respective eigenvalue's magnitude, we can glean across which axes we have relative noise (small eigenvalues) vs. important data (large eigenvalues).

Problem 4

a) YTM Plot

For plot of Canadian bond YTM please turn to Section GraphI in the appendix. I interpolated between points with straight lines as I felt this made the least assumptions on the data between any two points. I don't really expect the real YTM to lie exactly in the middle of any two bonds; it would be difficult to conject whether it would be above or below this point.

b) Spot Rates

The algorithm for deriving spot rates is reasonably simple, for any day of data our bond maturing in six months is functionally a zero coupon bond, with notional equal to the sum of face value and coupon. We now have our observed price, a notional and time until stock matures, using the compounding formula, $r_1 = \left(\frac{F+C}{P}\right) - 1$, we can find our spot rate for our first bond. For our n'th bond we proceed inductively, assuming we have r_j for $j < n$. Rearranging our price formula, $P = \sum_{t=1}^{n-1} \frac{C}{(1+r_t)^t} + \frac{F+C}{(1+r_n)^n}$, we solve that $r_n = \left(\frac{F+C}{P - \sum_{t=1}^{n-1} \frac{C}{(1+r_t)^t}} - 1 \right)^{1/n}$. For a plot of spot rates using canadian bond please refer to Section GraphII in the appendix

c) Forward Rates

To calculate the forward rate we simply have to use the formula $F_{t,t+n} = \left(\frac{(1+S_{t+n})^{t+n}}{(1+s_t)^t} \right)^{\frac{1}{n}} - 1$. S_t is the spot rate for time t, in this problem t = 1. t + n therefore is the difference between the maturity dates of a bond maturing in one year and a bond maturing in n years. Looping over our spot rates for 2 to 5 (although I did 1.5 to 5 in 0.5 increments since we have the data) we get our desired forward rates. For a plot of forward rates using Canadian bond data please refer to Section GraphIII in the appendix

Problem 5

YTM Time Series Covariance Matrix

Please refer to Section MatrixI for the covariance matrix of the YTM time series.

Forward Rate Time Series Covariance Matrix

Please refer to Section MatrixII for the covariance matrix of the YTM time series.

Problem 6

YTM eigenvectors and eigenvalues

Please refer to Section EigenvaluesI for the covariance matrix of the YTM time series.

FR eigenvectors and eigenvalues

Please refer to Section EigenvaluesII for the covariance matrix of the YTM time series.

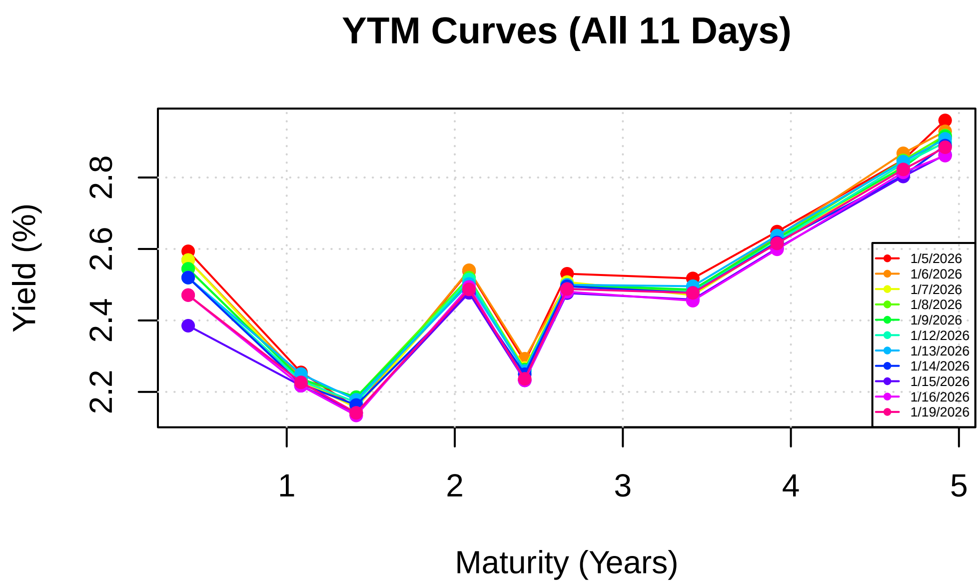
Comments

This vector in bond space points in the direction where the data is most spread out, that is the weighted combination of stocks that explains the most risk. The size of it's eigenvalue, compared to the others tells us how much of the variation is explained by this portfolio.

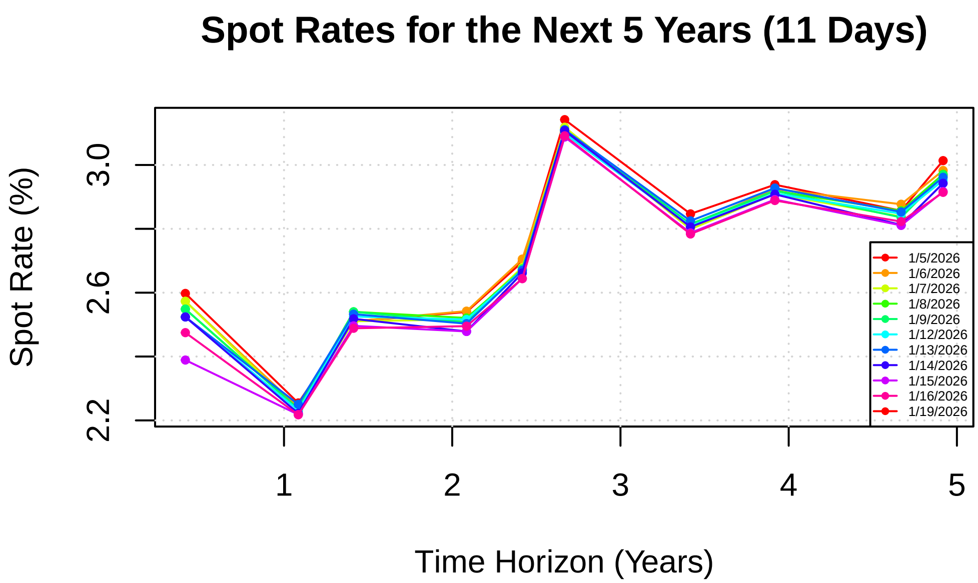
Assumptions, Tables, Graphs

It was assumed prices were quoted on the first of January, and that every year has 365 days. For discounting it was assumed that the final coupon payment made at the same date of maturity (together with face value).

Graph I

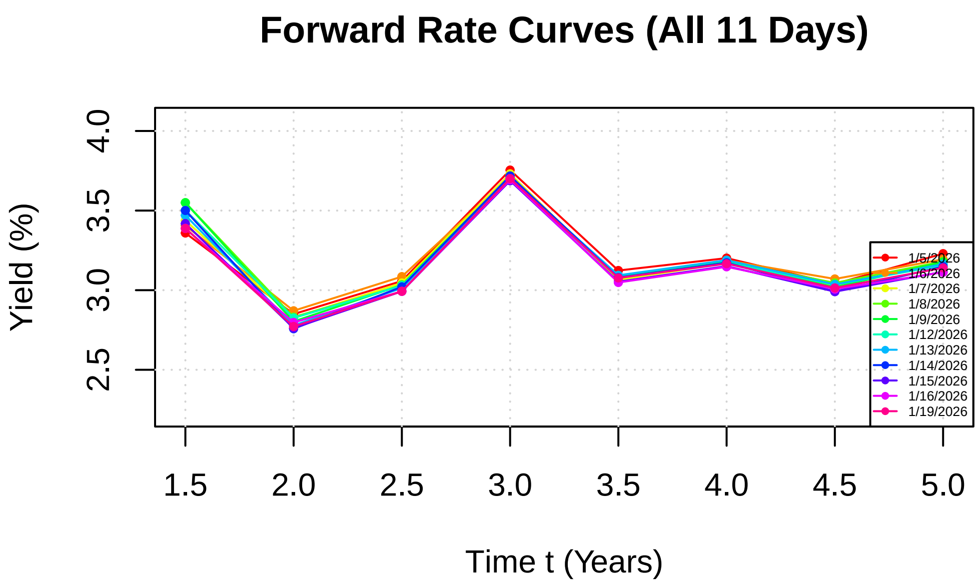


Graph II



Spot Rates for the Next 5 Years (11 Days)
Spot Rate (%)
3.0
2.6
2.2
1
2
3
4
5
Time Horizon (Years)
1/5/2026
1/6/2026
1/7/2026
1/8/2026
1/9/2026
1/12/2026
1/13/2026
1/14/2026
1/15/2026
1/16/2026
1/19/2026

Graph III



Matrix I

47.853	1.273	3.806	0.982	3.693	6.591	4.564	3.265	-0.119	5.260
1.273	3.318	2.027	0.861	0.758	1.416	3.321	2.659	2.366	2.616
3.806	2.027	4.525	0.089	0.791	0.756	2.670	2.471	1.740	2.496
0.982	0.861	0.089	3.025	1.032	-1.074	0.009	0.455	3.580	0.096
3.693	0.758	0.791	1.032	2.120	0.018	0.800	1.157	2.082	0.994
6.591	1.416	0.756	-1.074	0.018	2.585	2.621	1.461	-1.217	2.155
4.564	3.321	2.670	0.009	0.800	2.621	4.955	3.344	1.129	4.008
3.265	2.659	2.471	0.455	1.157	1.461	3.344	2.688	1.936	2.874
-0.119	2.366	1.740	3.580	2.082	-1.217	1.129	1.936	5.849	1.205
5.260	2.616	2.496	0.096	0.994	2.155	4.008	2.874	1.205	3.501

Matrix II

25.480	-1.658	1.782	-0.532	0.609	1.600	-0.132	1.683
-1.658	10.615	2.527	-3.441	-3.167	-1.425	6.352	-2.311
1.782	2.527	4.033	-0.610	-0.833	0.213	1.989	-0.085
-0.532	-3.441	-0.610	2.914	1.975	0.584	-3.402	1.658
0.609	-3.167	-0.833	1.975	4.959	2.869	-0.126	4.140
1.600	-1.425	0.213	0.584	2.869	2.207	1.475	2.682
-0.132	6.352	1.989	-3.402	-0.126	1.475	8.350	0.518
1.683	-2.311	-0.085	1.658	4.140	2.682	0.518	3.831

Eigenvalues I

The columns are the eigenvectors in increasing eigenvalues. The respective eigenvalue is listed across the top.

0.00051	0.00016	8.5e-05	2.8e-05	1.3e-05	6e-06	1.7e-06	5.3e-07	6e-08	1.4e-20
0.9539	0.2452	-0.1110	0.0135	0.0581	0.0322	0.0146	-0.0601	0.0332	0.0898
0.0586	-0.4107	0.0651	-0.2269	0.2372	0.6506	-0.2925	0.1971	0.2740	0.3052
0.1044	-0.3389	0.0775	0.8692	0.1509	0.0018	-0.1763	0.1140	0.0044	-0.2141
0.0215	-0.1491	-0.5083	-0.2322	0.3599	-0.3912	-0.4162	0.3563	-0.2677	-0.1033
0.0825	-0.1342	-0.2358	-0.0164	-0.8630	0.0744	-0.3524	0.1446	0.0681	-0.1385
0.1473	-0.0769	0.3609	-0.2849	0.0511	0.2292	0.1016	0.2057	-0.2352	-0.7732
0.1329	-0.4317	0.3216	-0.2218	0.0177	-0.4133	-0.3073	-0.5757	0.2014	-0.0761
0.0969	-0.3671	0.0919	-0.0018	-0.1508	0.0863	0.1440	-0.1241	-0.8127	0.3473
0.0171	-0.4070	-0.6121	-0.0321	0.0395	0.1475	0.5053	-0.3075	0.1561	-0.2467
0.1390	-0.3514	0.2162	-0.0851	-0.1265	-0.4057	0.4527	0.5594	0.2584	0.1948

Eigenvalues II

The columns are the eigenvectors in increasing eigenvalues. The respective eigenvalue is listed across the top.

0.00026	0.00019	0.00011	3.4e-05	2.3e-05	4.2e-06	8.4e-07	1.1e-07
0.9668	-0.1960	0.1211	0.0612	-0.0751	-0.0268	-0.0359	0.0293
-0.1671	-0.6710	-0.0427	-0.1290	-0.6809	0.0919	0.0127	-0.1763
0.0499	-0.2254	-0.0723	-0.8719	0.3846	0.1675	-0.0331	0.0657
0.0277	0.3117	-0.0391	-0.3618	-0.3737	-0.7299	-0.2941	0.1037
0.0892	0.2653	-0.5426	0.0020	-0.2638	0.5084	-0.4662	0.2861
0.0957	0.0742	-0.4137	0.0077	0.1405	-0.0784	-0.1590	-0.8735
-0.0554	-0.5052	-0.5069	0.2770	0.3524	-0.4024	-0.1524	0.3141
0.1182	0.1795	-0.5042	-0.1083	-0.1721	-0.0598	0.8032	0.0975

References & GitHub link

Github

Milstein, E., & Wessel, D. (n.d.). What did the Fed do in response to the COVID-19 crisis? Brookings. Retrieved January 14, 2026, from https://www.brookings.edu/articles/fed-response-to-covid19/

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